

ADITYA T. VADLAMANI

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EDUCATION

The Ohio State University

PhD in Computer Science & Engineering

M.S. in Computer Science & Engineering

B.S. in Computer Science & Engineering, Mathematics, Magna Cum Laude

Columbus, OH

Aug. 2023 – Present

Jan. 2022 – May 2023

Aug. 2018 – May 2022

EXPERIENCE

Machine Learning Engineer Intern

May 2026 – Aug. 2026

Pinterest

Remote

- Introduced a new architecture and training objectives of OmniSage, Pinterest's in-house representation learning framework, to better leverage user engagement signals
- Improved key offline evaluation metrics by up to 5% through the new architecture and objectives

Software Engineer Intern

May 2025 – Aug. 2025

Google

Sunnyvale, CA

- Designed and implemented a TF-GNN-based graph neural network to rank code tests most likely to fail when machine learning libraries are modified across Google's codebase
- Built a large-scale data pipeline using an internal Flume-based framework to construct code-test dependency graphs from Google's codebase for model training

Software Engineer Intern

May 2023 – Aug. 2023

Cruise LLC

Sunnyvale, CA

- Designed and implemented a prototype encoder model to detect invalid autonomous vehicle simulation scenarios
- Reduced manual review and accelerated software releases by filtering low-quality simulation outputs

Software Development Engineer Intern

May 2022 – Aug. 2022

Amazon Web Services

Cupertino, CA

- Prototyped a scalable AWS environment for functional network verification of routing policies.

RESEARCH EXPERIENCE

Graduate Student Researcher

Aug. 2022 – Present

The Ohio State University

Columbus, OH

- Developed a framework for fair conformal prediction, extended to federated learning (ICLR 2025, NeurIPS 2026).
- Benchmarked conformal prediction methods in transductive node classification (TMLR 2025).
- Developed scalable training methods for heterogeneous GNNs using sampling and ensembling (ICDM 2025).
- Proposed a framework for auditing AI systems in real-world deployments (KDD 2026, Blue Sky Ideas Track).
- Extending conformal prediction to generative models via diffusion steering and LLM safety guarantees.
- Administered multi-GPU HPC infrastructure for large-scale ML and data-intensive research.

SELECTED PUBLICATIONS

FedCF: Fair Federated Conformal Prediction, NeurIPS 2026. Acceptance Rate: 25.7%

Towards Auditing AI Systems in the Wild, KDD 2026. Blue Sky Ideas Track. Acceptance Rate: 23.61%.

A Generic Framework for Conformal Fairness, ICLR 2025. Acceptance Rate: 31.75%

Conformal Prediction: A Theoretical Note and Benchmarking Transductive Node Classification in Graphs, TMLR 2025

BLB-HGNN: Bag of Little Bootstraps for Training Heterogeneous GNNs, ICDM 2025. Acceptance Rate: 22.42%

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL, MATLAB, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, HuggingFace Transformers, NumPy, Pandas

Graph Machine Learning: PyTorch Geometric, DGL, TF-GNN, NetworkX

MLOps & Experiment Tracking: Docker, Weights & Biases, MLflow

Distributed Systems & HPC: Ray, SLURM, CUDA, Apache Spark, Apache Flume

Cloud Platforms & DevOps: AWS, Google Cloud Platform, Terraform, Git

PROJECTS

Online Store (Graduate Teaching Associate Project) | *Vue.js, Express.js, AWS, Terraform*: Scaled a full-stack application from a few pilot sections to all of OSU's first-year engineering program (2,000+ students).